Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Term Evaluation Theory(Even) Semester Regular Examination February 2026

Roll no...............................

Name of the Course and semester: MTech 2nd semester (CSE)
Name of the Paper: Deep Learning
Paper Code: MCS-201
Time: 1.5 Hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. **(10 Marks)**

a. What is Perceptron? Explain the historical context and development of biologically inspired computing. Describe the basic structure of a Perceptron with a neat diagram and explain how it mimics biological neurons. (CO1)

OR

b. Explain different activation functions used in deep learning. Write their mathematical equations and draw their graphs. (CO2)

Q2. **(10 Marks)**

a. Describe the architecture of Convolutional Neural Networks (CNN). Explain the role of Convolution Layer, Pooling Layer, and Fully Connected Layer with suitable diagrams. CO2)

OR

b. What is Transfer Learning? Explain the concepts of freezing input layers and fine-tuning output layers. Discuss the advantages of using transfer learning in deep learning applications with examples. (CO2)

Q3. **(10 Marks)**

a. Explain the Backpropagation algorithm in Multi-layer Perceptron. Consider a simple neural network with one hidden layer having 2 neurons. Given:

i. Input: $x = [1, 2]$
ii. Weights (input to hidden): $W_1 = [[0.5, 0.3], [0.4, 0.6]]$
iii. Weights (hidden to output): $W_2 = [0.7, 0.5]$
iv. Bias values: $b_1 = [0.1, 0.2]$, $b_2 = 0.3$
v. Target output: $y = 1$
vi. Activation function: Sigmoid

Perform:
i. Forward propagation to calculate output
ii. Calculate the loss using Mean Squared Error
iii. Explain backward propagation steps (no need to calculate complete gradients). (CO1)

OR

b. Discuss various regularization techniques used in deep learning and Data Augmentation. Explain how Data Augmentation techniques help prevent overfitting with examples. (CO2)

Q4. **(10 Marks)**

a. Explain Stochastic Gradient Descent (SGD) optimization algorithm. Compare it with Batch Gradient Descent and Mini-batch Gradient Descent. (CO1)

OR

b. What is the Bias-Variance tradeoff in deep learning? Explain with mathematical formulation how total error is decomposed into bias, variance, and irreducible error. Discuss techniques to reduce high bias and high variance in neural network models. (CO2)

Q5. **(10 Marks)**

a. Discuss the evolution of CNN architectures in the ImageNet challenge. Compare the following architectures, LeNet, AlexNet, VGG-16, Google Net, ResNet-50 in detail including their architectural innovations, Trade-offs between depth, parameters, and computational efficiency and how they addressed the vanishing gradient problem. (CO1)

OR

b.) Explain the problems of vanishing and exploding gradients and how proper weight initialization helps to overcome them in deep neural networks. (CO2)